CS-502 Fundamentals Of Algorithms

Update MCQS For Quiz-2 File

Solve By Vu Topper RM

85% To 100% Marks



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The Dynamic Programming Edit Distance Algorithm takes a total running time of _____.
A. Θ (nlogn)
B. Θ (n3)
C. Θ (n)
D.Θ (n2)

In Ram, each instruction involves performing some basic operation on how many values in the machines memory?
A. four
B. one
C. three
D. two

Which of the following is calculated with Big Omega notation?
A. Medium bounds
B. Upper bounds
C. Lower bounds
D. Both upper and lower bounds

Asymptotic growth of Plane-sweep algorithm for 2-D maxima problem is _____.
A. Θ(logn)
B. Θ(nlogn)
C. Θ(n)
D. Θ(n^2)

Two functions 4n2 and (8n2 + 2n − 3) are _____.
A. related to power functions
B. hard to calculate
C. asymptotically equivalent
D. not equivalent

The O-notation is used to state only the asymptotic _____bounds.
A. Upper
B. Lower
C. Both lower & upper
D. Two

Which one of the following is worst case time complexity of merge sort?
A. O((n^2logn)
B. O(nlogn)
C. O(nlogn^2)
D. O(n^2)

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The important factors to measure the running time of the brute-force 2-d maxima algorithm are

A. All of above

B. The number of comparisons that are performed

C. Count the number of times an element of P is accessed

D. Count the number of steps of the pseudo code that are executed

To predict the cost of an algorithm in terms of resources is called _____.

A. Output

B. Analysis

C. Design

D. Key

The total no. of arguments passed to merge sort algorithm is _____.

A. 5

B. 3

C. 2

D. 4

Plane-Sweep uses _____ for storing maximal points.

A. queue

B. stack

C. array

D. list

A Random Access Machine (RAM)is an idealized machine with _____ random access memory.

A. Infinite large

B. 2 GBs

C. 256 MB

D. 512 MB

RAM is idealized machine with _____ large random-access memory.

A. Average

B. Infinite

C. Finite

D. Small

Θ(1) means _____.

A. time can not be calculated

B. that if used for small n

C. that it is used for large n

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D. constant amount of time

if A dominates B, and B dominates C, then it is necessary that
A. C dominates B
B. A dominates C
C. B dominates A
D. C dominates A

An algorithm is a mathematical entity, which is independent of _____.
A. Programming language
B. Programming language, Compiler and Machine
C. Machine and Programming language
D. Compiler and Programming language

The _____ is the weighted sum of running times with weights.
A. best-case time
B. average-case time
C. both best-case and average-case.
D. worse-case time

There are no hard formal rules to the syntax of the _____ code.
A. Assembly
B. Basic
C. Programming
D. Pseudo

For n > 1, _____ divides into two halves, sorts the two and then combine them together.
A. Heap Sort
B. Insertion Sort
C. MergeSort
D. Quick Sort

Plane-Sweep algorithm can run in _____ time.
A. Θ(n log)
B. Θ(n)
C. Θ(n*n)
D. Θ(n log n)

If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a _____ car.
A. Expensive
B. Cheap

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C. Slow
D. Fast

Heap sort is a/an _____ and _____ sorting algorithm.

A. Not in-place, not stable one
B. In-place, not stable one
C. Not in-place, stable one
D. In-place, stable one

In Bucket sort, if there are duplicates then each bin can be replaced by a _____.

A. Hash table
B. Heap
C. Stack
D. Linked list

If there are duplicates in _____ sort, then each bin can be replaced by a linked list.

A. Heap
B. Merge
C. Bucket ok
D. Quick

A sorting algorithm is called as _____ if duplicate elements remain in the same relative position after sorting.

A. O(n) algorithm
B. Complex
C. Parallel
D. Stable

The average case running time of quick sort algorithm is theta _____.

A. n x n log(n)
B. Log(n)
C. nlog(n) ok
D. (n)

Quick sort algorithm required a lot of comparisons in the _____ condition.

A. Worst case
B. Best and Average case
C. Average case
D. Best case

Memoization is a part of _____ programming strategy.

A. dynamic

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B. fast
C. memory
D. slow

_____ programming is essentially recursion without repetition.

A. Array ok
B. Dynamic
C. N log n
D. fast

In _____, the size of recursive calls is totally depending on how pivot is chosen.

A. Quick sort ok
B. Merge sort
C. Insert sort
D. Bubble sort

The main shortcoming of counting sort is that it is useful for_____.

A. large integers
B. large real numbers
C. small real numbers
D. small integers ok

Comparison-based sorting algorithms always takes _____ time.

A. Theta nlog(n) ok
B. Omega n(n^2)
C. Big Oh nlog(n)
D. Omega nlog(n)

Quick sort does not require any additional array for storage except for recursive function calls is called _____.

A. In-Place ok
B. Stable
C. Not In-Place
D. Unsorted

While solving Selection problem, in Sieve technique we choose pivot _____

A. Minimum element
B. Randomly
C. Average element
D. Maximum element

While applying the sieve technique, _____ subarray will contain all elements that are greater than pivot element x.

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A. A[q+1....n]
B. A[1....n]
C. A[1....q-1]
D. A[q]

The total running time for Selection algorithm is _____ in n.
A. Exponential
B. Geometric
C. Quadratic
D. Linear

What is the worst-case time of a quick sort which happens rarely?
A. O(log n)
B. O(n log n)
C. O(n2) **ok**
D. O(n)

While applying the sieve technique, _____ subarray will contain all elements that are less than pivot element x.
A. A[q+1....n]
B. A[1....q-1]
C. A[1....n]
D. A[q]

A Principal operation for maintaining the heap property is called heapify, it is also called:
A. Sifting Up
B. Sifting left
C. Sifting right
D. Sifting down **Page 43**

After partitioning array in Quick sort, pivot is placed in a position such that
A. Values larger than pivot are on left and smaller than pivot are on right
B. Pivot is the first element of array
C. Pivot is the last element of array
D. Values smaller than pivot are on left and larger than pivot are on right **Page 35**

Sieve technique is a special case of _____ strategy.
A. Greedy approach
B. Graph
C. Divide-and-Conquer
D. Dynamic programming

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Selection sort takes theta _____ in the worst case.

A. (n)
B. (n^2)
C. nlog(n)
D. n(logn)

Array divided into _____ subarrays while applying sieve technique to selection problem.

A. 1
B. 2
C. 3
D. 4

If input “n” is odd, then median will be _____

A. n/2
B. n+2
C. (n-1)/2
D. (n+1)/2 **Page 34**

Quick sort is based on _____ strategy.

A. Graph Theory
B. Greedy approach
C. Divide-and-Conquer **ok**
D. Dynamic programming

_____ is one of the few problems, where provable lower bounds exist on how fast we can sort.

A. Sorting **Page 39**
B. Graphing
C. Searching
D. Both Searching & Sorting

In Quick sort algorithm, the subarray _____ has elements which are less than pivot element x.

A. A[q]
B. A[p....r]
C. A[q+1....r]
D. A[p....q-1] **ok**

In _____ sorting algorithm, we just need to swap positions of data during the Partitioning function.

A. Merge sort
B. Counting sort
C. Radix sort
D. Quick sort

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Question No: 01 **(Marks:1)** **Vu-Topper RM**

There are _____ entries in the Edit Distance Matrix.

A. Θ (n)
B. Θ (n2) **Page 84** **ok**
C. Θ (n+2)
D. Θ (n + 100)

Question No: 02 **(Marks:1)** **Vu-Topper RM**

For average-case time analysis of quick sort algorithm, pivot selection is on average basis from _____.

A. All possible random values **Page 50** **ok**
B. Pivot is input separately
C. half of the input values
D. Values greater than 5

Question No: 03 **(Marks:1)** **Vu-Topper RM**

As per algorithm of dynamic programing, we need to store the result(s) of _____.

A. First sub-problem only
B. Best solution only
C. Intermediate sub-problems **Page 75** **ok**
D. Final solution only

Question No: 04 **(Marks:1)** **Vu-Topper RM**

In chain matrix multiplication, table is filled _____ to find the multiplication of matrix.

A. row wise
B. column wise
C. diagonally
D. bottom-to-up **Page 86**

Question No: 05 **(Marks:1)** **Vu-Topper RM**

The only way to convert a string of i characters into the empty string is with i deletions, represented as _____.

A. E(0.j) =j
B. E(i.j) = 1
C. E(0.i) = j
D. E (i.0)=i **Page 78** **ok**

Question No: 06 **(Marks:1)** **Vu-Topper RM**

If there are Θ (n^2)entries in edit distance matrix then the total running time is:

A. θ (n)
B. θ (1)
C. θ (n^2) **Page 84** **ok**
D. θ (n logn)

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Question No: 07 **(Marks:1)** **Vu-Topper RM**

In average –case time analysis of quick sort algorithm , the most balanced case for partion is when we divide the list of elements into _.

A. Equal no. of pieces as of input elements
B. Single piece exactly
C. Two nearly equal pieces
D. Three nearly equal pieces

Question No: 08 **(Marks:1)** **Vu-Topper RM**

If matrix A of dimension p x q is multiplied with matrix B of dimension q x r, then each entry in resultant matrix takes _____ time.

A. O (q) **Page 84**
B. (1)
C. (p x q)
D. (q x r)

Question No: 09 **(Marks:1)** **Vu-Topper RM**

Fibonacci Sequence was named on _____, a famous mathematician in 12th Century.

A. Fred Brooks
B. Grady Booch
C. Leonardo Pisano **Page 73** **ok**
D. Edgar F. Codd

Question No: 10 **(Marks:1)** **Vu-Topper RM**

In quick sort algorithm, we choose pivot _____.

A. Always the smallest element
B. Greater than 5
C. Randomly **Page 35** **ok**
D. Less than 5

Question No: 11 **(Marks:1)** **Vu-Topper RM**

For comparison-based sorting algorithms, it is _____ possible to sort more efficiently than Omega nlog(n) time.

A. Always
B. Sometimes not
C. NOT **Page 54** **ok**
D. Sometimes

Question No:12 **(Marks:1)** **Vu-Topper RM**

The sequence of merge sort algorithm is:

A. Divide Combine-Conquer
B. Conquer-Divide-Combine
C. Divide-Conquer-Combine **Page 27**
D. Combine-Divide-Conquer

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Question No: 13 **(Marks:1)** **Vu-Topper RM**

In _____ Knapsack Problem, limitation is that an item can either be put in the bag or not. Fractional items are not allowed.

A. 0
B. 1
C. 0/1 **Page 91**
D. Fractional

Question No: 14 **(Marks:1)** **Vu-Topper RM**

In Selection algorithm, we assume pivot selection takes theta _____ running time.

A. n **Page 36**
B. n2
C. n3
D. log (n)

Question No: 15 **(Marks:1)** **Vu-Topper RM**

In Heap Sort algorithm (using max heap), when every time maximum element is removed from top _____.

A. We call merge Sort Algorithm
B. it becomes Order n2 Algorithm
C. Divide and Conquer strategy helps us
D. We are left with a hole **Page 41** **ok**

Question No: 16 **(Marks:1)** **Vu-Topper RM**

_____ is a method of solving a problem in which we check all possible solutions to the problem to find the solution we need.

A. Plane-Sweep Algorithm
B. Sorting Algorithm
C. Brute-Force Algorithm **Google** **ok**
D. Greedy approach

Question No: 17 **(Marks:1)** **Vu-Topper RM**

The worst case running time of quick sort algorithm _____.

A. Is quadratic
B. Is linear
C. Cannot be quadratic
D. ls always Exponential

Question No: 18 **(Marks:1)** **Vu-Topper RM**

In max heap (for Heap Sort algorithm), when every time maximum element is removed from top we replace it with _____ leaf in the tree.

A. Last **Page 41**
B. First

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C. Any
D. Second last

Question No: 19 (Marks:1) Vu-Topper RM

Quick sort algorithm was developed by –

A. AlferdAho
B. Sedgewick
C. John Vincent Atanasoff
D. Tony Hoare Google

Question No: 20 (Marks:1) Vu-Topper RM

If Matrix-A has dimensions "3×2" and Matrix-B has dimensions "2×3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

A. 3×2
B. 2×3
C. 2×2
D. 3×3

Question No: 21 (Marks:1) Vu-Topper RM

In Sorting the key value or attribute _____ from an ordered domain.

A. Must be Page 39
B. Not always
C. May be
D. Occasionally

Question No: 22 (Marks:1) Vu-Topper RM

Result of asymptotical analysis of n(n -3) and 4n*n is that _____

A. n(n-1) is asymptotically Less
B. n(n-1) is asymptotically Greater
C. Both are asymptotically Not equivalent
D. Both are asymptotically Equivalent Page 23

Question No: 23 (Marks:1) Vu-Topper RM

Floor and ceiling are _____ to calculate while analyzing algorithms

A. Very easy
B. 3rd Option is missing
C. Usually considered difficult
D. 4th Option is missing

Question No: 24 (Marks:1) Vu-Topper RM

_____ of reference is an important fact of current processor technology.

A. Defining
B. Assigning
C. Locality Page 8 ok

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D. Formality

Question No: 25 (Marks:1) Vu-Topper RM

In max-heap, largest element is stored at root node. Where is the smallest element stored?

A. Right Node
B. Leaf Node Google
C. Middle Node
D. Left Node

Question No: 26 (Marks:1) Vu-Topper RM

Which of the following is calculated with Big O notation?

A. Medium bounds
B. Upper bounds
C. Lower bounds Page 25 ok
D. Both upper and lower bounds

Question No: 27 (Marks:1) Vu-Topper RM

Edit distance algorithm based on_____strategy.

A. Greedy
B. Dynamic Programming Page 81 ok
C. Divide and Conquer
D. Searching

Question No: 28 (Marks:1) Vu-Topper RM

In Heapsort Algorithm, total time taken by heapify procedure is ____

A. O (log n) Page 43
B. (log2 n)
C. (n log n)
D. (n2 log n)

Question No: 29 (Marks:1) Vu-Topper RM

Al-Khwarizmi was a/an _____
Artist

A. Astronomer
B. Mathematication Page 7
C. Khalifah

Question No: 30 (Marks:1) Vu-Topper RM

When matrix A of 5 x 3 is multiplied with matrix B of 3 x 4 then the number of multiplications required will be _____.

A. 15
B. 12
C. 36
D. 60 Page 84

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Question No: 31 **(Marks:1)** **Vu-Topper RM**

Pseudo code of algorithms are to be read by _____.

A. People **Page 12** **ok**

B. RAM

C. Computer

D. Compiler

Question No: 32 **(Marks:1)** **Vu-Topper RM**

The sieve technique is a special case, where the number of sub-problems is just _____

A. 1 **Page 34**

B. 2

C. 3

D. 4

Question No: 33 **(Marks:1)** **Vu-Topper RM**

When a recursive algorithm revisits the same problem over and over again, we say that the optimization problem has _____ sub-problems.

A. Overlapping **Google** **ok**

B. Over costing

C. Optimized

D. Three

Question No: 34 **(Marks:1)** **Vu-Topper RM**

In order to say anything meaningful about our algorithms, it will be important for us to settle on a _____.

A. Java Program

B. C++ Program

C. Pseudo program

D. Mathematically model of computation **ok**

Question No: 35 **(Marks:1)** **Vu-Topper RM**

Merge sort is based on_____.

A. Brute-force

B. Plan-sweep

C. Axis-sweep

D. Divide and Conquer **ok**

Question No: 36 **(Marks:1)** **Vu-Topper RM**

What time does Merge Sort algorithm take in order to sort an array of 'n' numbers?

A. Θ (n)

B. Θ (log n)

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C. Θ (n^2)
D. Θ(n log n) **Page 30** **ok**

Question No: 37 **(Marks:1)** **Vu-Topper RM**
algorithm, the first step is to _____.
A. Call Build-Heap procedure **Page 46**
B. Sort the array in descending order
C. Call Heapify procedure
D. Find the number of input elements

Question No: 38 **(Marks:1)** **Vu-Topper RM**
The definition of theta-notation relies on proving _____ asymptotic bound.
A. One
B. Lower
C. Upper
D. Both lower & upper **Page 25**

Question No: 39 **(Marks:1)** **Vu-Topper RM**
In merge sort algorithm, to merge two lists of size n/2 to a list of size n, takes _____ time.
A. Theta (n) **Page 32**
B. Theta log(n)
C. Theta log2(n)
D. Theta n log(n)

Question No: 40 **(Marks:1)** **Vu-Topper RM**
We can make _____ recursive calls in Fibonacci Sequence.
A. Infinite
B. Finite **Google** **ok**
C. Only one
D. Zero

Question No: 41 **(Marks:1)** **Vu-Topper RM**
Following is NOT the application of Edit Distance problem.
A. Speech recognition
B. Spelling Correction
C. Ascending Sort **Page 76** **ok**
D. Computational Molecular Biology

Question No: 42 **(Marks:1)** **Vu-Topper RM**
In plane sweep approach, a vertical line is swept across the 2d-plane and structure is used for holding the maximal points lying to the left of the sweep line.
A. Tree
B. Array

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C. Queue
D. Stack

Question No: 43 (Marks:1) Vu-Topper RM
Time will vary according to the nature of input data.
_____ time is the maximum running time over all legal inputs.
A. Worst-case Page 13
B. Average-case
C. Best-case
D. Good-case

Question No: 44 (Marks:1) Vu-Topper RM
Efficient algorithm requires less computational…
A. Memory
B. Running Time
C. Memory and Running Time Page 9
D. Energy

Question No: 45 (Marks:1) Vu-Topper RM
Selection algorithm takes theta _____
A. (n2)
B. (n)
C. log(n)
D. n log(n)

Question No: 46 (Marks:1) Vu-Topper RM
Time complexity of Dynamic Programming based algorithm for computing the minimum cost of Chain Matrix Multiplication is _____
A. Log n
B. n
C. n^2 (n square)
D. n^3 (n cube) Page 90

Question No: 47 (Marks:1) Vu-Topper RM
The Iteration method is used for _____
A. Solving Recurrence relations Page 31
B. Merging elements in Merge sort
C. Comparing sorting algorithms only
D. Dividing elements in Merge sort

Question No: 48 (Marks:1) Vu-Topper RM
In 3-Dimensional space, a point P has _____ coordinate(s).
A. (X, Y)
B. (X, 0)
C. (0, Y)
D. (X,Y, Z) ok

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Question No: 49 **(Marks:1)** **Vu-Topper RM**

Chain matrix multiplication problem can be solved through _____ strategy.

A. Dynamic programming **Page 85**

B. Greedy

C. Divide and conquer

D. Sorting

Question No: 50 **(Marks:1)** **Vu-Topper RM**

Merge sort have running time….running time of Heap sort. Not found exactly

A. Greater than

B. Less than **Google**

C. Equal to

D. Different than

Question No: 51 **(Marks:1)** **Vu-Topper RM**

The Omega-notation allows us to state only the asymptotic _____bounds.

A. Middle

B. Lower **Page 25** **ok**

C. Upper

D. Both lower & upper

Question No: 52 **(Marks:1)** **Vu-Topper RM**

Both lower &upperSorting can be in _____

A. Random order

B. Increasing order only

C. Decreasing order only

D. Both Increasing and Decreasing order

Question No: 53 **(Marks:1)** **Vu-Topper RM**

Quicksort is a/an _____ and _____ sorting algorithm.

A. Not in place, not stable one

B. In place , not stable one **Page 54** **ok**

C. In place , stable one

D. Not in place , stable one

Question No: 54 **(Marks:1)** **Vu-Topper RM**

Consider three matrices X,Y,Z of dimensions 1×2, 2×3,3×4 respectively. The number of multiplications of (XY) Z is:

A. 18

B. 32

C. 24

D. 30

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Question No: 55 **(Marks:1)** **Vu-Topper RM**

In Dynamic Programming, our approach is to _____.

A. Express the problem non-recursively

B. Build the solution in a bottom-up fashion **Page 75** **ok**

C. Develop the solution in a top-down fashion

D. Input several sub-problems simultaneously

Question No: 56 **(Marks:1)** **Vu-Topper RM**

The knapsack problem is optimally solved by using brute force algorithm.Counting sort is suitable to sort the elements in range 1 to K;

A. K is large

B. K is small **Page 57**

C. K may be large or small

D. None

Question No: 57 **(Marks:1)** **Vu-Topper RM**

Matrix multiplication is a(n) _____ operation.

A. Commutative

B. Associative **Page 85**

C. Neither commutative nor associative

D. Commutative but not associative

Question No: 58 **(Marks:1)** **Vu-Topper RM**

In Dynamic Programming approach, solution is modified/changed _____.

A. Always once

B. At each stage **Google** **ok**

C. Only for specific problems

D. At 4th stage only

Question No: 59 **(Marks:1)** **Vu-Topper RM**

In Knapsack problem, the goal is to put items in the Knapsack such that the value of the items is _____ subject to weight limit of knapsack.

A. Minimized

B. Decreased

C. Maximized **Page 91**

D. None of the given options

Question No: 60 **(Marks:1)** **Vu-Topper RM**

An in-place sorting algorithm is one that _____ use(s) additional array for storage.

A. Always

B. Permanently

C. Does not **Page 54** **ok**

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D. Sometime

Question No: 61 (Marks:1) Vu-Topper RM

Dynamic Programming is a problem-solving approach in which___

A. Problem is solved in Zero time
B. Solution is developed only at final stage
C. Both are correct
D. Both are incorrect Google

Question No: 62 (Marks:1) Vu-Topper RM

In Fibonacci Sequence, each term is calculated by_____ previous _____ terms.

A. Subtracting, Two
B. Adding, Three
C. Adding, Two Page 73 ok
D. Multiplying, Two

Question No: 63 (Marks:1) Vu-Topper RM

Dynamic programming formulation of the matrix chain multiplication problem will store the solutions of each sub problem in a(n):

A. Class
B. Array
C. Table
D. Variable

Question No: 64 (Marks:1) Vu-Topper RM

Sorting is performed on the basis of _____.

A. Computational resources
B. Asymptotic notation
C. Summation
D. Some key value of attribute Page 39 ok

Question No: 65 (Marks:1) Vu-Topper RM

In Heap Sort algorithm, we call Build-heap procedure _____.

A. Twice
B. Thrice
C. Only once Page 46 ok
D. As many times as we need

Question No: 66 (Marks:1) Vu-Topper RM

In the statement “output P[1].x, P[1].y”, the number of times elements of P are accessed is _____.

A. 1
B. 2 Page 14 ok
C. 3
D. 4

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Question No: 67 **(Marks:1)** **Vu-Topper RM**

_____ provides us more accurate result, when input values are not closer with each other.

A. Mode
B. Mean
C. Average
D. Median **Page 34**

Question No: 68 **(Marks:1)** **Vu-Topper RM**

The process of _____ ends when you are left with such tiny pieces remaining that it is trivial to solve them.

A. Brute-force
B. Plan-sweep
C. Axis-sweep
D. Divide and Conquer **ok**

Question No: 69 **(Marks:1)** **Vu-Topper RM**

Rank of an element can be defined as _____

A. One minus the number of elements that are smaller
B. Two plus the number of elements that are greater
C. One plus the number of elements that are smaller **Page 34**
D. Two minus the number of elements that are smaller

Question No: 70 **(Marks:1)** **Vu-Topper RM**

If the time complexity of an algorithm is given by O (1), then its time complexity would be

A. Polynomial
B. Exponential
C. Constant **Google**
D. Average

Question No: 71 **(Marks:1)** **Vu-Topper RM**

The asymptotic growth of n(n+1)/2 is:

A. O(n)
B. $O(n^2)$ **ok**
C. O(n+2)
D. O(n log n)

Question No: 72 **(Marks:1)** **Vu-Topper RM**

Approach of solving geometric problems by sweeping a line across the plane is called _____sweep.

A. Line
B. Plane **Page 18** **ok**
C. Cube

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D. Box

Question No: 73 (Marks:1) Vu-Topper RM

In Sieve technique, we solve the problem _____

A. In recursive manner Page 34

B. Non recursively

C. Using Merge Sort algorithm

D. Using Brute force technique

Question No: 74 (Marks:1) Vu-Topper RM

One of the limitation in 0/1 knapsack is that an item can either be ____ in the bag or not.

A. Use

B. Put Page 91

C. Move

D. Store

Question No: 75 (Marks:1) Vu-Topper RM

Which one is not passed as parameter in Quick sort algorithm?

A. End of the array

B. Start of the array

C. Middle of the array

D. Array (containing input elements) Google ok

Question No: 76 (Marks:1) Vu-Topper RM

In the analysis of Selection algorithm, we get the convergent _____ series.

A. Harmonic

B. Linear

C. Arithmetic

D. Geometric Page 37

Question No: 77 (Marks:1) Vu-Topper RM

A Random Access Machine (RAM)is an idealized machine withrandom access memory.

A. Infinite large Page 10

B. 512 MB

C. 256 MB

D. 2 GBs

Question No: 78 (Marks:1) Vu-Topper RM

While analyzing Selection algorithm, we make a number of passes, in fact it could be as many as _____.

A. n(n+1)

B. log(n) Page 37 ok

C. n/3

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D. n/4

Question No: 79 (Marks:1) Vu-Topper RM

In Random Access Machine, Instructions are executed

A. Parallel
B. Batch
C. One by One Page 10 ok
D. Multiple times

Question No: 80 (Marks:1) Vu-Topper RM

In selection problem, the rank of an element will be its _____ position

A. First
B. final Page 34
C. Second last
D. Last

Question No: 81 (Marks:1) Vu-Topper RM

The worst-case running time of merge sort is _____ in order to sort an array of n elements.

A. O(log n)
B. O(n)
C. O(n log n) Page 40 ok
D. O(n)

Question No: 82 (Marks:1) Vu-Topper RM

f(n) and g(n) are asymptotically equivalent. This means that they have essentially the same _____.

A. Size
B. Results
C. Variables
D. Growth Rates

Question No: 83 (Marks:1) Vu-Topper RM

An algorithm is a mathematical entity. Which is independent of ____.

A. Programming language
B. Machine and Programming language
C. Compiler and Programming language
D. Programing Language Compiler and Machine

Question No: 84 (Marks:1) Vu-Topper RM

In Quick sort algorithm, Pivots form ___

A. Stack
B. Queue
C. Graph
D. Binary Search Tree Page 49

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Vu Topper RM

Question No: 85 **(Marks:1)** **Vu-Topper RM**

Counting sort is suitable for sorting the elements within range 1 to P, where _____

A. P is large
B. P is Small ok
C. P is very large
D. P is undetermined

Question No: 86 **(Marks:1)** **Vu-Topper RM**

In asymptotical analysis of n*(5 + 2) – 3 , as n becomes large, the dominant (fastest growing) term is some constant times _____

A. n_1
B. n
C. n+1
D. n*n Page 23 ok

Question No: 87 **(Marks:1)** **Vu-Topper RM**

___ Items are not allowed in the 0/1 knapsack.

A. Lighter
B. Whole
C. Weighty
D. Fractional

Question No: 88 **(Marks:1)** **Vu-Topper RM**

In partition algorithm, the subarray _____ has elements which are greater than pivot element x.

A. A[q]
B. A[p…r]
C. A[p…q-1]
D. A[q+1…r] Page 46

Question No: 89 **(Marks:1)** **Vu-Topper RM**

In Heap Sort algorithm, if heap property is violated:

A. We ignore.
B. We call Heapify procedure Page 43
C. We call Build Heap procedure.
D. Heap property can never be violated.

Question No: 90 **(Marks:1)** **Vu-Topper RM**

_____ is not a characteristic of Random Access Machine.

A. Assigning a value to a variable
B. Locality of reference
C. Single-Processor Page 10 ok
D. Executing an arithmetic instruction

Vu Topper RM

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Question No: 91 **(Marks:1)** **Vu-Topper RM**

The only way to convert an empty string into a string of j characters is by doing j insertions, represented as _____.

A. E(i,j) = 1
B. E(I,0) = I
C. E(0,j) = j **Page 78** **ok**
D. E(1,j)= j

Question No: 92 **(Marks:1)** **Vu-Topper RM**

In Selection problem, the Sieve technique works in _____

A. Non-recursive manner
B. Constant time
C. Phases **Page 34** **ok**
D. One complete go

Question No: 93 **(Marks:1)** **Vu-Topper RM**

Algorithm is a sequence of computational steps that _____ the input into output.

A. Merge
B. Assign
C. Transform **Page 7** **ok**
D. Integrate

Question No: 94 **(Marks:1)** **Vu-Topper RM**

If pj dominates pi and pi dominates ph then pj also dominates ph, it means dominance relation is

A. Transitive **Page 18**
B. Non Transitive
C. Equation
D. Symbolic

Question No: 95 **(Marks:1)** **Vu-Topper RM**

To find maximal points in brute-force algorithm each point of the space is compared against _____ of that space.

A. One other point
B. All other points **Page 11**
C. Few other points
D. Most of the other points

Question No: 96 **(Marks:1)** **Vu-Topper RM**

In the following code the statement "cout<<j;" executes ---------- times.for (j=1; j<=5; j = j+2)
cout<<j;

A. 5 times
B. 2 times
C. 3 times **ok**

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D. 0 times

Question No: 97 **(Marks:1)** **Vu-Topper RM**

In merge sort algorithm, we split the array _____ to find index q.

A. Midway **Page 17** **ok**

B. Exiting

C. Entring

D. Summing

Question No: 98 **(Marks:1)** **Vu-Topper RM**

In Selection problem, the Sieve technique _____

A. Add some more input items each time

B. Do not work recursively

C. Do not uses Divide and Conquer approach

D. Eliminates undesired data items each time **Page 35**

Question No: 99 **(Marks:1)** **Vu-Topper RM**

Consider three matrices X, Y, Z of dimensions 1 x 2 , 2 x 3 , 3 x 4 respectively. The number of multiplications of (XY)Z is:

A. 16

B. 32 **Page 84**

C. 30

D. 26

Question No:100 **(Marks:1)** **Vu-Topper RM**

In Heap Sort algorithm, the total running time for Heapify procedure is _____.

A. Theta (log n) **Page 43** **ok**

B. Order (log n)

C. Omega (log n)

D. O(1) i.e. Constant time

Question No:101 **(Marks:1)** **Vu-Topper RM**

The sieve technique works where we have to find _____ item(s) from a large input.

A. Single **Page 34**

B. Two

C. Three

D. Similar

Question No:102 **(Marks:1)** **Vu-Topper RM**

In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object i , then we gain_____

A. W(Total Weight of Knapsack)

B. V (Total Value of all items)

C. vi (Value of object i) **Page 93**

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D. None of the given option

Question No:103 (Marks:1) Vu-Topper RM

While Sorting, the order domain means for any two input elements x and y__ satisfies only.

A. x < y Page 39

B. x > y

C. x = y

D. All of the above

Question No:104 (Marks:1) Vu-Topper RM

For solving Selection problem, we introduced Sieve technique due to _____

A. Using Decrease and Conquer strategy Page 34 ok

B. Avoiding to sort all input data

C. Eliminating Rank of an element

D. Using Brute-force approach

Question No:105 (Marks:1) Vu-Topper RM

In quick sort algorithm, _____ decides nature of Binary Search Tree formed by pivots.

A. Rank of the pivot Page 49 ok

B. Middle element from input

C. Smallest element from input

D. Largest element from input

Question No:106 (Marks:1) Vu-Topper RM

In plane sweep approach, a vertical line is swept across the 2d-plane from_____.

A. Right to Left

B. Left to Right Page 18 ok

C. Top to Bottom

D. Bottom to top

Question No:107 (Marks:1) Vu-Topper RM

For small values of n, _____ algorithm is fast enough.

A. Medium

B. Large

C. Small Page 14 ok

D. Infinity

Question No:108 (Marks:1) Vu-Topper RM

Dynamic programming comprises of _____.

A. Recursion only

B. Repetition only

C. Recursion with Repetition

D. No Repetition but Recursion Page 75

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Question No:109 **(Marks:1)** **Vu-Topper RM**

The function f(n)=n(logn+1)/2 is asymptotically equalient t nlogn :
Here Lower Bound means function f(n) grows asymptotically at __ as fast as nlog n.

A. Least **Page 23**
B. Normal
C. Most
D. AT

Question No:110 **(Marks:1)** **Vu-Topper RM**

Counting sort has time complexity.

A. O(n+k)
B. O(n) **Page 58**
C. O(k)
D. O(nlogn)

Question No:111 **(Marks:1)** **Vu-Topper RM**

Due to left complete nature of binary tree, the heap can be stored in

A. Array **Page 40**
B. Structures
C. Link List
D. Stack

Question No:112 **(Marks:1)** **Vu-Topper RM**

Single item from a larger set of _____.

A. Constant
B. Pointers
C. Phases
D. n items **Page 34**

Question No:113 **(Marks:1)** **Vu-Topper RM**

In the clique cover problem, for two vertices to be in the same group, they must be _____ each other.

A. Apart from
B. Far from
C. Near to
D. Adjacent to **Page 76**

Question No:114 **(Marks:1)** **Vu-Topper RM**

How much time merge sort takes for an array of numbers?

A. T(n^2)
B. T(n)
C. T(log n)

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D. T(n log n) **Page 40**

Question No:115 **(Marks:1)** **Vu-Topper RM**

In in-place sorting algorithm is one that uses arrays for storage.

A. No additional array **Page 54**

B. An additional array

C. Both of above may be true according to algorithm

D. More than 3 arrays of one dimension

Question No:116 **(Marks:1)** **Vu-Topper RM**

Brute-force algorithm for 2D-Maxima is operated by comparing _____ pairs of points.

A. Two

B. Some

C. Most

D. All **Page 18**

Question No:117 **(Marks:1)** **Vu-Topper RM**

While Sorting, the ordered domain means for any two input elements x and y _____ satisfies only.

A. x > y

B. x < y

C. x = y

D. All of the above **Page 38**

Question No:118 **(Marks:1)** **Vu-Topper RM**

Quick sort is.

A. Not stable but in place **Page 54**

B. Stable but not in place

C. Stable & in Place

D. Some time stable & some times in place

Question No:119 **(Marks:1)** **Vu-Topper RM**

Which may be a stable sort?

A. Merger

B. Insertion

C. Both above **Page 54**

D. None of the above

Question No:120 **(Marks:1)** **Vu-Topper RM**

For the Sieve Technique we take time.

A. T(nk) **Page 34**

B. IT(n / 3)

C. n^2

D. n/

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Question No:121 **(Marks:1)** **Vu-Topper RM**

Continuation sort is suitable to sort the elements in range 1 to k.

A. K is Large
B. K is not known
C. K may be small or large
D. K is small **Page 54**

Question No:122 **(Marks:1)** **Vu-Topper RM**

Asymptotic growth rate of the function is taken over _____ case running time. .

A. Worst **Page 14**
B. Average
C. Best
D. Normal

Question No:123 **(Marks:1)** **Vu-Topper RM**

Before sweeping a vertical line in plane sweep approach, in start sorting of the points is done in increasing order of their _____ coordinates. .

A. Y
B. Z
C. X
D. X , Y

Question No:124 **(Marks:1)** **Vu-Topper RM**

In Quick sort, we don't have the control over the sizes of recursive calls.

A. True **Page 49**
B. False
C. Less information to decide
D. Ether true or false

Question No:125 **(Marks:1)** **Vu-Topper RM**

Random access machine or RAM is a/an.

A. Machine build by Al-Khwarizmi
B. Mechanical machine
C. Mathematical model **Page 10**
D. Electronics machine

Question No:126 **(Marks:1)** **Vu-Topper RM**

A heap is a left-complete binary tree that confirms to the _____.

A. increasing order only
B. decreasing order only
C. heap order **Page 40**

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D. log n order

Question No:127 **(Marks:1)** **Vu-Topper RM**

Which one of the following sorting algorithms is the fastest?

A. Merge sort
B. Quick sort ok
C. Insertion sort
D. Heap sort

Question No:128 **(Marks:1)** **Vu-Topper RM**

Quick sort algorithm divide the entire array into _____ sub arrays.

A. 2 ok
B. 3
C. 4
D. 5

Question No:129 **(Marks:1)** **Vu-Topper RM**

In brute force algorithm, we measure running time T(n) based on_____.

A. Worst-case time and best-case time
B. Worst-case time and average-case time Page 46 ok
C. Average-case time and best-case time
D. Best-case time and staring-case time

Question No:130 **(Marks:1)** **Vu-Topper RM**

algorithm first of all _____.

A. Sorts all points
B. Delete some points
C. Output the elements
D. Pushes all points on stack

Question No:131 **(Marks:1)** **Vu-Topper RM**

Which symbol is used for Omega notation?

A. (O)
B. (Θ)
C. (Ω)
D. (@)

Question No:132 **(Marks:1)** **Vu-Topper RM**

Selection sort is a _____sorting algorithm.

A. In-place Page 54 ok
B. Not In-Place
C. Stable
D. in-partition

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Question No:133 **(Marks:1)** **Vu-Topper RM**

We do not need to prove comparison-based sorting algorithms by mathematically. It always takes _____ time.

A. Big Oh nlog(n)
B. Omega nlog(n)
C. Omega n(n^2)
D. Theta nlog(n)

Question No:134 **(Marks:1)** **Vu-Topper RM**

Merge sort is a/an _____ and _____ sorting algorithm.

A. Not in-place, not stable one
B. In-place, not stable one
C. In-place, stable one
D. Not in-place, stable one **Page 54** **ok**

Question No:135 **(Marks:1)** **Vu-Topper RM**

Cubic function will _____ a quadratic function.

A. Prove
B. Be equal to
C. Overtake **Page 25**
D. Find

Question No:136 **(Marks:1)** **Vu-Topper RM**

Insertion sort is a _____sorting algorithm.

A. Unstable
B. In-place **Page 54** **ok**
C. Not In-Place
D. in-partition

Question No:137 **(Marks:1)** **Vu-Topper RM**

To check whether a function grows faster or slower than the other function, we use some asymptotic notations, which is _____.

A. Big-oh notation
B. Theta notation **ok**
C. Omega notation
D. All of the given

Question No:138 **(Marks:1)** **Vu-Topper RM**

Asymptotic growth of 8n^2 + 2n – 3 is:

A. Θ(n^2 + n)
B. Θ (n^2) **Page 14**
C. Θ(8n^2)
D. Θ(8n^2 + 2n)

Question No:139 **(Marks:1)** **Vu-Topper RM**

In the analysis of algorithms,_____ plays an important role.

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A. Time **ok**
B. Money
C. Growth rate
D. Text analysis

Question No:140 **(Marks:1)** **Vu-Topper RM**
In inductive approach of knapsack problem, we consider 2 cases, ____ Or _____.
A. Median, Mode
B. Recursive, Iterative
C. Leave object, Take object **Page 93**
D. Sequentially. Parallel

Question No:141 **(Marks:1)** **Vu-Topper RM**
Random Access Machine (RAM) can execute _____ instructions.
A. Parallel
B. Only logical
C. Only arithmetic
D. Logical and arithmetic **ok**

Question No:142 **(Marks:1)** **Vu-Topper RM**
Using _____ algorithm, efficiency is not given much importance.
A. Greedy
B. Merge sort
C. Processing **ok**
D. Brute Force

Question No:143 **(Marks:1)** **Vu-Topper RM**
Bubble sort takes theta _____ in the worst case
A. (n2) **Page 39**
B. (n)
C. log(n)
D. nlog(n)

Question No:144 **(Marks:1)** **Vu-Topper RM**
Using base condition we set all m[i,i] = _____ ?
A. 1
B. 0 **Page 86**
C. ∞
D. -1

Question No:145 **(Marks:1)** **Vu-Topper RM**
Dynamic Programming algorithms often use some kind of _____ to store the results of intermediate sub-problems.

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A. Stack
B. Loop
C. Table ok
D. variable

Question No:146 (Marks:1) Vu-Topper RM
_____is in-place sorting algorithm.
A. Bubble sort Page 54 ok
B. Merge sort
C. Linear search
D. Binary Search

Question No:147 (Marks:1) Vu-Topper RM
Which one of the following problems can be solved using dynamic problem?
A. Bubble sort problem
B. Greedy search problem
C. Fractional knapsack problem
D. Matrix chain multiplication problem Page 85

Question No:148 (Marks:1) Vu-Topper RM
In chain matrix multiplication, solutions of the sub-problems are stored in a_____.
A. Array
B. Table Page 86
C. Tree
D. Link list

Question No:149 (Marks:1) Vu-Topper RM
What is the average running time of a quick sort algorithm?
A. O(n^2)
B. O(n)
C. O(n log n) Page 49 ok
D. O(log n)

Question No:150 (Marks:1) Vu-Topper RM
Sorting Algorithms having O _____ running time are considered to be slow ones.
A. (n)
B. (n^2) Page 39 ok
C. (nlog(n))
D. (log(n))

Question No:151 (Marks:1) Vu-Topper RM
While solving Selection problem, in Sieve technique we partition input data _____

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A. Randomly
B. According to Pivot Page 35
C. In increasing order
D. In decreasing order

Question No:152 (Marks:1) Vu-Topper RM
_____ is the process of avoiding unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later.
A. Loop
B. Function
C. Recursion
D. Memoization Page 74 ok

Question No:153 (Marks:1) Vu-Topper RM
In average-case time the probability of seeing input is denoted by ____.
A. p{I}
B. p[I]
C. p<i>
D. p(i) Page 13

Question No:154 (Marks:1) Vu-Topper RM
While applying the Sieve technique to selection sort, how to choose a pivot element.
A. Through mean
B. Linear
C. Randomly Page 35
D. Sequentially

Question No:155 (Marks:1) Vu-Topper RM
Number of _____of the pseudo code are counted to measure the running time.
A. Inputs
B. Outputs
C. Steps Page 13 ok
D. Pages

Question No:156 (Marks:1) Vu-Topper RM
Developing a dynamic programming algorithm generally involves _____ separate steps.
A. One
B. Two Page 75 ok
C. Three
D. Four

Question No:157 (Marks:1) Vu-Topper RM

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8n^2+2n+3 will exceed c28(n), no matter how large we make _____.

A. n
B. 2n
C. c2 Page 25
D. this quadratic equation

Question No:158 (Marks:1) Vu-Topper RM

The running time of quick sort algorithm _____.

A. Is impossible to compute
B. Has nothing to do with pivot selection
C. Is Random upon each execution
D. Is Greatly influenced by the selection of pivot Page 49

Question No:159 (Marks:1) Vu-Topper RM

_____ involves breaking up the problem into sub problems whose solutions can be combined to solve the global problem.

A. Complexity Theory
B. Greedy Algorithms
C. Divide and Conquer Strategy Page 34
D. Dynamic programming solution

Question No:160 (Marks:1) Vu-Topper RM

In _____ we have to find rank of an element from given input.

A. Merge sort algorithm
B. Selection problem Page 34
C. Brute force technique
D. Plane Sweep algorithm

Question No:161 (Marks:1) Vu-Topper RM

How many steps are involved to design the dynamic programming strategy?

A. 2
B. 4 Page 92
C. 3
D. 1

Question No:162 (Marks:1) Vu-Topper RM

In bin sort, each bin can be replaced by a _____ in case of duplication.

A. Heap
B. Stack
C. Hash table
D. Linked list Page 69 ok

Question No:163 (Marks:1) Vu-Topper RM

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Vu Topper RM

Question No:164 **(Marks:1)** **Vu-Topper RM**

Find the maximum value of the items which can carry using knapsack
Knapsack weight capacity = 50.
Item Weight Value
11070
22020
33080
470 200

A. 90
B. 280
C. 200
D. 100

Question No:165 **(Marks:1)** **Vu-Topper RM**

In 2-D maxima problem, A point p is said to be dominated by point q if _____.

A. p.x<= q.x
B. p.x<= q.x and p.y<= q.y **Page 17** **ok**
C. p.y<= q.y
D. p.x>= q.x and p.y>=q.y

Question No:166 **(Marks:1)** **Vu-Topper RM**

Sorting can be in _____.

A. Increasing order only
B. Decreasing order only
C. Both increasing and decreasing order **Page 39** **ok**
D. Random order

Question No:167 **(Marks:1)** **Vu-Topper RM**

Recurrence can be described in terms of_____.

A. Array
B. Linear
C. Tree **Page 31** **ok**
D. Graph

Question No:168 **(Marks:1)** **Vu-Topper RM**

The brute-force algorithm for 2D-Maxima runs in order O(___) time.

A. n
B. n(log n)
C. n*n **Page 18** **ok**
D. n3

Question No:169 **(Marks:1)** **Vu-Topper RM**

In plane sweep approach of solving geometric problems, a _____is swept across the plane.

A. Line **Page 18** **ok**

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B. Plane
C. Cube
D. Box

Question No:170 (Marks:1) Vu-Topper RM

Question No:171 (Marks:1) Vu-Topper RM

_____ is always based on divide and conquer strategy.

A. Bubble sort
B. Selection sort
C. Pigeon sort
D. Quick sort Page 46

Question No:172 (Marks:1) Vu-Topper RM

If a matrix has three rows and two columns, then dimensions of matrix will be:

A. 3×2
B. 2×3
C. 3×3
D. 2×2

Question No:173 (Marks:1) Vu-Topper RM

Asymptotic notations are used to describe _____ of an algorithm.

A. Size
B. Length
C. Running time Google ok
D. Compile time

Question No:174 (Marks:1) Vu-Topper RM

Catalan numbers are related the number of different _____ on 'n' nodes.

A. Arrays
B. linked lists
C. binary trees Page 85
D. functions

Question No:175 (Marks:1) Vu-Topper RM

Applying the sieve technique to selection problem, _____element is picked from array.

A. Pivot Page 35 ok
B. Total
C. Input
D. Output

Question No:176 (Marks:1) Vu-Topper RM

In recursive formulation of knapsack Problem: V [0, j] = ____ for j>=0

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A. 2
B. -1 Page 93
C. 1
D. 2

Question No:177 (Marks:1) Vu-Topper RM
_____ is a linear time sorting algorithm.
A. Merge sort
B. Radix sort Page 71 ok
C. Quick sort
D. Bubble sort

Question No:178 (Marks:1) Vu-Topper RM
Quick sort is one of the _____sorting algorithm.
A. Fastest Page 19
B. Slowest
C. Major
D. Average

Question No:179 (Marks:1) Vu-Topper RM
The time assumed for each basic operation to execute on RAM model of computation is _____.
A. Infinite
B. Continuous
C. Constant Page 10 ok
D. Variable

Question No:180 (Marks:1) Vu-Topper RM
While analyzing algorithms, _____and _____ are usually considered difficult to calculate.
A. Floor, ceiling Google
B. Row, Column
C. Finite, Infinite
D. Graph, Tree

Question No:181 (Marks:1) Vu-Topper RM
While analysis of the brute-force maxima algorithm, an array sorted in the reverse order is the type of _____ case input.
A. Best
B. Worst Page 14
C. Somewhat bad
D. Average

Question No:182 (Marks:1) Vu-Topper RM
_____ is not useful measure of central tendency of given input set

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especially when the distribution of values is highly skewed.

A. Mean
B. Mode
C. Average
D. Median Page 34

Question No:183 (Marks:1) Vu-Topper RM

In asymptotical analysis of n(n - 3) and 4n*n, as n becomes large, the dominant (fastest growing) term is some constant times_____.

A. n+1
B. n-1
C. n
D. n*n Page 23 ok

Question No:184 (Marks:1) Vu-Topper RM

In addition to passing in the array itself to merge sort algorithm, we will pass in _____other arguments which are indices.

A. Three
B. Two ok
C. Four
D. Five

Question No:185 (Marks:1) Vu-Topper RM

In 2d-maximal problem, a point is said to be if it is not dominated by any other point in that space.

A. Member
B. Minimal
C. Maximal
D. Joint

Question No:186 (Marks:1) Vu-Topper RM

Counting sort assumes that the numbers to be sorted are in the range _____.

A. K to n where n is large
B. K to n where k is small
C. 1 to k where k is small ok
D. k to n where n is small

Question No:187 (Marks:1) Vu-Topper RM

Insertion sort is an efficient algorithm for sorting a _____ number of elements.

A. Small
B. Large Page 39
C. Extra large
D. Medium

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Question No:188 **(Marks:1)** **Vu-Topper RM**

If the indices passed to merge sort algorithm are _____,then this means that there is only one element to sort.

A. Small **Page 28**

B. Large

C. Equal

D. Not Equal

Question No:189 **(Marks:1)** **Vu-Topper RM**

In Knapsack Problem, each item must be entirely accepted or rejected, is called _____ problem.

A. Linear

B. Fractional

C. 0-1

D. Optimal

Question No:190 **(Marks:1)** **Vu-Topper RM**

If the time complexity of an algorithm is O(n), then it is called _____ time complexity.

A. Linear **ok**

B. Constant

C. Average

D. Exponential

Question No:191 **(Marks:1)** **Vu-Topper RM**

In the case of _____, analysis does not depend upon on the distribution of input.

A. Merge sort

B. Insertion sort

C. Quick Sort **ok**

D. Heap sort

Question No:192 **(Marks:1)** **Vu-Topper RM**

We can use the _____ property to devise a recursive formulation of the edit distance problem.

A. Small substructure

B. Algorithmic

C. Real

D. Optimal substructure **Page 78** **ok**

Question No:193 **(Marks:1)** **Vu-Topper RM**

The following sequence is called _____.1,2,3,5,8,13,21,34,55,…..

A. Fibonacci sequence **Page 73** **ok**

B. Optimal sequence

C. Optimize Sequence

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D. Overlapping sequence

Question No:194 (Marks:1) Vu-Topper RM
Which one sorting algorithm is best suited to sort an array of 2 million elements?

A. Insert sort
B. Bubble Sort Page 71 ok
C. Merge sort
D. Quick sort

Question No:195 (Marks:1) Vu-Topper RM
We can improve the performance of quick sort if we could be able to _____.

A. Select two or more pivots Page 34 ok
B. Skip any sub-array completely
C. Skit Input elements somehow
D. Eliminate recursive calls

Question No:196 (Marks:1) Vu-Topper RM
The problem with the brute-force algorithm is that it uses _____ in pruning out decisions.

A. Worst-case time
B. No intelligence Page 18 ok
C. Outside looping
D. Artificial intelligence

Question No:197 (Marks:1) Vu-Topper RM
In Heap Sort algorithm, Heapify procedure is _____ in nature.

A. Recursive Page 43 ok
B. Non-Recursive
C. Fast
D. Slow

Question No:198 (Marks:1) Vu-Topper RM
An algorithm is said to be correct if for every _____ instance, it halts with the correct _____.

A. Input, Output Page 13
B. Design, Analysis
C. Value, Key
D. Key, Analysis

Question No:199 (Marks:1) Vu-Topper RM
If we have an equation 8n2+7f*n + 5f + 6 then is large, _____ term will be muchxxxxxxxthe n term and will dominate the running time.

A. f g (n)
B. g (n) * 2

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C. n * 2 **Page 23**
D. f (n)

Question No:200 **(Marks:1)** **Vu-Topper RM**
For quick sort algorithm, partitioning takes theta _____.
A. (n)
B. log(n)
C. n log (n)
D. n2log (n) **Google**

Question No:201 **(Marks:1)** **Vu-Topper RM**
In Heap Sort algorithm, the maximum levels an element can move upward is _____.
A. Theta (log n) **Page 43** **ok**
B. Big-ch (log n)
C. Omega (log n)
D. 0 (1) i.e. Constant time

Question No:202 **(Marks:1)** **Vu-Topper RM**
Which process is used for avoiding unnecessary repetitions and looking them up again if we need them later.
A. Greedy Approach
B. Memoization **Page 74** **ok**
C. Divide and conquer
D. Recursion

Question No:203 **(Marks:1)** **Vu-Topper RM**
The worst-case running time of Quick sort is _____ in order to sort an array of n element.
A. O(n log n) **Page 49**
B. O(n)
C. $O(n^2)$
D. O(log n)

Question No:204 **(Marks:1)** **Vu-Topper RM**
Boolean operation is a _____ operation on an idealized RAM model of computation.
A. Advance
B. String
C. Basic
D. Normal

Question No:205 **(Marks:1)** **Vu-Topper RM**
In chain matrix multiplication, if there are n items, there are _____ ways in which outer most pair of parentheses can placed.
A. n^2

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B. 2n
C. n+1
D. n-1 Page 85

Question No:206 (Marks:1) Vu-Topper RM
The number of nodes in a complete binary tree of height h is:
A. (h+1) – 1
B. (h+1)
C. 2^(h+1) – 1 Page 40
D. ((h+1)^2) – 1

Question No:207 (Marks:1) Vu-Topper RM
In Sieve Technique, we know the item of interest.
A. True
B. False

Question No:208 (Marks:1) Vu-Topper RM
The Huffman codes provide a method of encoding data inefficiently when coded using ASCII standard.
A. True
B. False Page 99

Question No:209 (Marks:1) Vu-Topper RM
In Heap Sort algorithm, we build _____ for ascending sort.
A. Min heap
B. Max Heap Page 41
C. Both
D. None of these

Question No:210 (Marks:1) Vu-Topper RM
Quick sort is a recursive algorithm.
A. True
B. False

Question No:211 (Marks:1) Vu-Topper RM
In Heap Sort algorithm, to remove the maximum element every time,_____.
A. Nothing happens
B. We call heapify procedure Google ok
C. We call Build-Heap procedure
D. Heap Sort algorithm terminates without result

Question No:212 (Marks:1) Vu-Topper RM
When a heapify procedure is applied to the root node to restore the heap, then at each level, the comparison performed takes time:
A. It will take O(1). Page 43 ok

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B. It will take Θ(log n).
C. It can not be predicted.
D. Time will vary according to the nature of input data.

Question No:213 (Marks:1) Vu-Topper RM
What is the best case time complexity of merge sort?
A. O((n^2)
B. O((nlogn) Google ok
C. O((nlogn^2)
D. O((n^2logn)

Question No:214 (Marks:1) Vu-Topper RM
In Heap Sort algorithm, the first step is to _____.
A. Call Heapify procedure
B. Call Build-Heap procedure Page 46
C. Sort the array in descending order
D. Find the number of input elements

Question No:215 (Marks:1) Vu-Topper RM
Merge sort algorithm discussed in handouts contains _____.
A. 1 loop
B. 3 loops
C. 2 loops Google ok
D. 4 loops

Question No:216 (Marks:1) Vu-Topper RM
In _____, Leonardo of Pisa, also called Fibonacci, published a book.
A. 1102
B. 1202 Google ok
C. 1400
D. 1304

Question No:217 (Marks:1) Vu-Topper RM
If matrix A of dimension 2 x 4 is multiplied with matrix B of dimension 4 x 3, then the dimension of resultant matrix will be _____.
A. 2x3 Page 84
B. 4x3
C. 3x4
D. 2x4

Question No:218 (Marks:1) Vu-Topper RM
In generating Fibonacci Sequence, we can avoid unnecessary repetitions by _____ process.
A. Loop
B. Function

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C. Recursion
D. Memoization Page 43 ok

Question No:219 (Marks:1) Vu-Topper RM
Algorithms similar to those for the _____ problem are used in some speech recognition systems.
A. Counting
B. heap sort
C. Fibonacci
D. edit-distance Page 77 ok

Question No:220 (Marks:1) Vu-Topper RM
Radix sort performs sorting the numbers _____ digit(s) at a time.
A. One Page 71 ok
B. Two
C. All
D. Four

Question No:221 (Marks:1) Vu-Topper RM
Radix sort is a _____ integer sorting algorithm.
A. In-Place
B. Unstable
C. Comparative
D. Non-comparative Google ok

Question No:222 (Marks:1) Vu-Topper RM
We can use the optimal substructure property to devise a _____ formulation of the edit distance problem.
A. Iterative
B. Optimum
C. Selective
D. Recursive Page 78 ok

Question No:223 (Marks:1) Vu-Topper RM
The formula for calculating the Catalan number is _____.

○	$C(n) = \frac{1}{n+1}\binom{n}{2n}$
○	$C(n) = \frac{1}{n-1}\binom{n}{2n}$
◉	$C(n) = \frac{1}{n+1}\binom{2n}{n}$ **Page 85**
○	$C(n) = \frac{1}{n-1}\binom{2n}{n}$

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Question No:224 **(Marks:1)** **Vu-Topper RM**

_____ belongs to Dynamic programming.

A. Heap sort
B. Merge sort
C. Edit distance Page 77 ok
D. Divide and conquer

Question No:225 **(Marks:1)** **Vu-Topper RM**

In his book _____, Leonardo Pisano addressed the Fibonacci sequence as well as a variety of other problems.

A. Liber fib
B. Fib abaci
C. Fibonacci
D. Liber abaci Google ok

Question No:226 **(Marks:1)** **Vu-Topper RM**

Dynamic Programming approach is usually useful in solving _____ problems.

A. Loop
B. Array
C. Normal
D. Optimization Page 97 ok

Question No:227 **(Marks:1)** **Vu-Topper RM**

We can multiply two matrices A and B only when they are compatible which means:

A. Number of rows and columns do not matter
B. Number of rows in A must be equal to number of rows in B
C. Number of columns in A must be equal to number of rows in B Page 84
D. Number of columns in A must be equal to number of columns in B

Question No:228 **(Marks:1)** **Vu-Topper RM**

If we have 6 metrices in chain matrix multiplication problem then the number of table entries must be?

A. 12
B. 25
C. 30
D. 36 Google

Question No:229 **(Marks:1)** **Vu-Topper RM**

_____ algorithm based on Dynamic Programming strategy.

A. Quick Sort

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B. Heap Sort **Google** **ok**
C. Binary Tree
D. Edit distance

Question No:230 **(Marks:1)** **Vu-Topper RM**
Which method is preferable for dealing with chain matrix multiplication?
A. Graph Theory
B. Greedy Approach
C. Divide and Conquer Strategy
D. Dynamic Programming Formulation **Google**

Question No:231 **(Marks:1)** **Vu-Topper RM**
_____ overcomes the limitations of _____ by working as per positional notations of numbers.
A. Bubble sort, Radix sort
B. Radix sort, Bubble sort,
C. Counting sort, Radix sort
D. Radix sort, Counting sort **Page 71** **ok**

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